Ground based light curve follow-up validation observations of TESS object of interest TOI-5516.01

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Abstract

NASA's Transiting Exoplanet Survey Satellite (TESS) has opened new opportunities for the discovery of exoplanets by continuously monitoring bright dwarf stars for transit signatures. Candidates identified by TESS, known as TESS Objects of Interest (TOIs), require ground-based follow-up to distinguish genuine planets from false positives. In this work, we report on follow-up observations of TOI 5516.01 using the 0.8 m telescope at George Mason University. The objective was to verify whether a transit could be

detected at the predicted location, time, and depth consistent with TESS observations. The data were processed and analyzed with AstroImageJ, incorporating reduction, plate solving, aperture and multi-aperture photometry, and nearby eclipsing binary checks to construct the light curve. The signal-to-noise ratio was insufficient to clearly distinguish a transit event from variability and noise in the data. As a result, the planetary status of TOI 5516.01 remains unresolved, and additional high-precision follow-up will be required.

1. Introduction

Exoplanetary science has advanced rapidly since the early 1990s, driven in large part by space-based surveys designed to detect planets around nearby stars. Among these, NASA's Transiting Exoplanet Survey Satellite (TESS) has been especially prolific, identifying thousands of planet candidates through the detection of periodic decreases in stellar brightness as potential planets transit their host stars [10]. These candidates, designated as TESS Objects of Interest (TOIs), represent a vast resource for understanding planetary demographics [1].

Yet, the raw detections from TESS alone cannot confirm the planetary nature of these signals. Transit-like features can be mimicked by astrophysical false positives such as eclipsing binaries, blended background stars, or instrumental systematics. As a result, ground-based photometric follow-up is an indispensable step in validating TOIs. By obtaining high-precision light curves at predicted transit times, astronomers can test whether a transit occurs at the correct location, depth, and duration [9,10]. Without such confirmation, the status of many TOIs remains uncertain; indeed, although TESS has identified more than 7,000 potential exoplanets, only a small fraction has been validated [4,5].

Photometric follow-up is typically carried out using modest-aperture telescopes paired with specialized reduction and analysis pipelines. AstroImageJ has become a widely adopted platform for this work, offering image calibration, plate solving, and single- and multi-aperture photometry, as well as diagnostic tools for identifying nearby eclipsing binaries. When combined with ephemeris data from the NASA Exoplanet Archive and ExoFOP, these methods allow for rigorous comparison between predicted and observed light curves [2].

The target of this study, TOI-5516.01, was flagged by TESS as a candidate planet with a predicted transit of shallow depth and relatively short duration, making it a challenging case for confirmation [12]. Its host star is bright enough to permit ground-based follow-up using mid-sized telescopes, yet the expected signal lies close to the detection threshold, raising the possibility of confusion with noise or nearby eclipsing sources. Confirming or refuting such candidates is essential not only for validating individual systems but also for refining occurrence rates of small exoplanets in the TESS catalog.

In this work, we report on follow-up observations of TOI-5516.01 conducted with George Mason University's 0.8 m telescope [11]. Our goal is to test whether a transit event can be detected at the predicted star, with the expected timing, duration, and depth derived from TESS data. The subsequent sections describe the observational campaign (Section 2), the photometric analyses of both TESS and ground-based data (Section 3), the resulting light curves (Section 4), and a discussion of our findings and limitations (Section 5), followed by conclusions and directions for future follow-up (Section 6).

2. Observations

2.1 TESS Observational Data

TOI 5516.01 (TIC 70678429) was flagged as a planet candidate by TESS in 2022 [12]. Table 1 summarizes the stellar and planetary parameters as reported in the TESS Input Catalog and candidate files.

Table 1. TOI 5516.01 system parameters from TESS

Parameter	Value	Notes
TIC ID	70678429	TESS Input Catalog
RA (J2000)	10h 23m 21.20s	—
Dec (J2000)	+13° 28′ 20.03″	—
Orbital period	~4.38 days	Transit fit
Transit midpoint	2459574.485 BJD	TESS
Transit duration	~2.86 hr	TESS
Transit depth	~3.9 ppt	TESS
Planet radius	~12.34 $R_\oplus$	Derived
Insolation flux	~448.1 $S_\oplus$	Derived
Equilibrium temperature	~1281 K	Derived
Stellar effective temperature	~6077 K	TIC
Stellar log g	~3.882 cm s^{-2}	TIC
Stellar radius	~2.025 $R_\odot$	TIC
Distance	~857 pc	TIC

These values classify TOI 5516.01 as a large-radius candidate with a shallow transit depth, placing it near the limits of ground-based detectability.

2.2 GMU Telescope Observations

Ground-based follow-up was conducted using the George Mason University (GMU) 0.8 m telescope in Fairfax, Virginia, during the predicted transit window. The observational setup and data collection are summarized in Table 2.

Table 2. GMU 0.8 m telescope observation log for TOI 5516.01

Parameter	**Value**
Telescope	GMU 0.8 m
Location	Fairfax, Virginia
Observation date	Feb 25–26, 2024
Observation window	23:55–10:48 UTC
Number of science frames	167
Exposure time	75 s
Filter	R-band

These observations were specifically designed to capture the predicted transit event and generate a light curve for comparison with TESS data [14].

We observed TOI-5516 on February 25, 2024, with the 0.8 m GMU telescope using a red filter. A total of 167 science images were obtained with 75s exposures.

3. Analysis

3.1 Tool

AstroImageJ (AIJ) was used to perform calibration, aperture photometry, multi-aperture photometry, and light curve generation [2, 15]. AIJ also provided diagnostic tools, such as the differential magnitude versus RMS (dmag–RMS) plot, to assess potential nearby eclipsing binaries (NEBs).

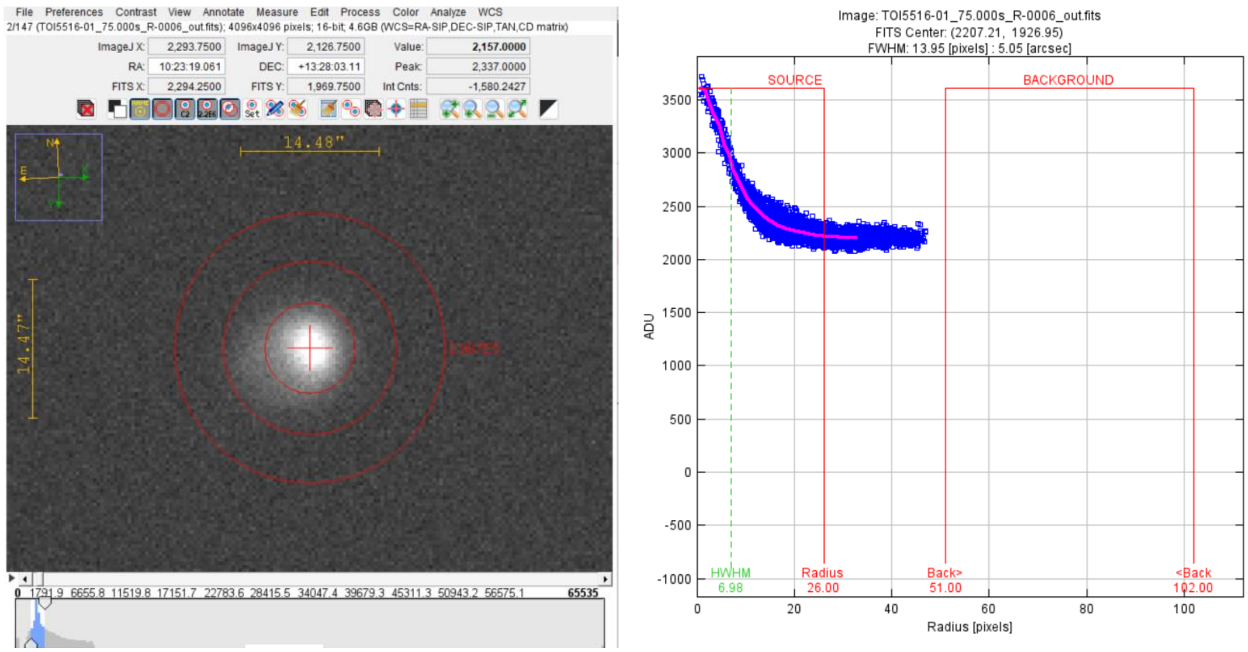
3.2 Data Reduction and Photometry in AstroImageJ

Science frames were reduced using AIJ's CCD Data Processor. Calibration involved creating a master flat field (3.5 s exposure) and two master dark frames (3.5 s and 75 s exposures) to match the flat and science frames, respectively. Raw science images were dark-subtracted and flat-divided, producing reduced science frames [6,7].

From the original 167 raw frames, 148 were retained after discarding frames affected by poor tracking, satellite streaks, or defocus. The reduced images were then plate-solved and aligned using the target coordinates of TOI-5516.01 to ensure consistency across the image stack.

3.3 Aperture Photometry and Reference Star Selection

Aperture photometry was performed on the target star, with an aperture radius of 13 pixels, inner annulus radius of 25 pixels, and outer annulus radius of 39 pixels (Figure 1(a)). These dimensions were guided by the curve shown in Figure 1(b).



(a) (b)

Figure 1. (a) Aperture photometry setup for TOI-5516.01 using *AstroImageJ*. The target star is centered, with the chosen source aperture (inner circle) and background annuli (outer circles) shown in red. (b) Radial profile of the target star, displaying the flux distribution in ADU versus radius (pixels). The selected source radius and background annuli are indicated, with the FWHM of the stellar profile measured at ~6.98 pixels (~5.05 arcsec).

To obtain a reliable transit light curve, we employed differential photometry, a technique that measures the relative brightness of a target star against nearby reference stars. This method corrects the systematic effects such as changing atmospheric transparency or instrumental drift that affect all stars in the field equally. In practice, we used AstroImageJ's multi-aperture photometry tool, placing apertures on both the target and carefully selected up to 15 reference stars. The target's aperture radius and the inner and outer radius of the background annulus were optimized to capture the stellar flux while minimizing sky noise. Reference stars were chosen to be of comparable brightness and apparent size to the target, ensuring robust normalization and reducing systematic errors in the extracted light curve (Figure 2). Due to limited nearby stars of matching brightness, some larger stars were included as references.

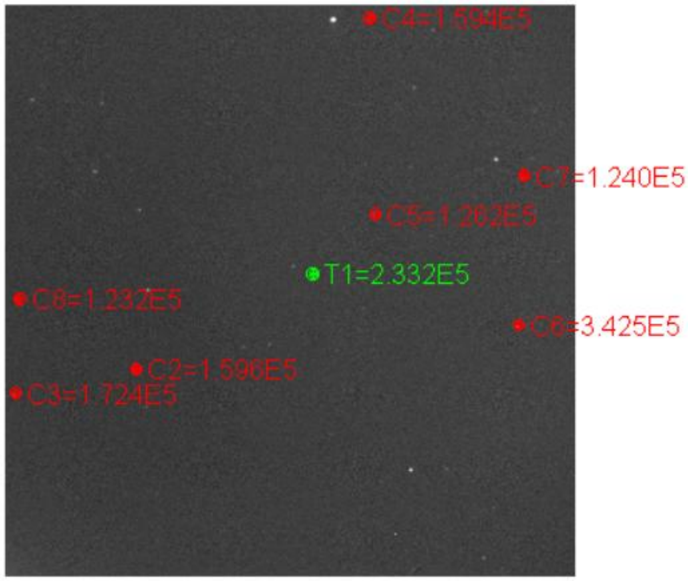


Figure 2. Selection of the target star and comparison stars in AstroImageJ. The target star (T1) is shown in green with a measured flux of 2.332 × 10^5 ADU. Eight comparison stars (C2–C8) are shown in red with their corresponding flux values. These stars were chosen based on brightness and proximity to the target to serve as reference stars for differential photometry.

3.4 Light Curve Modeling

To generate a ground-based light curve, stellar parameters (radius, metallicity, log g, Teff) from table 1 were input into AIJ's Data Set 2 Fit Settings window. Limb-darkening coefficients adopted were Linear LD u1 = 0.3 and Quadratic LD u2 = 0.3. Time stamps were converted from UTC to BJD, yielding a predicted ingress time of 0.611 BJD and egress time of 0.7302 BJD. To generate the light curve, we reviewed each reference star and discarded those that caused significant scattering or variation in the light curve. All the comparison stars were in green [6,15].

3.5 Nearby Eclipsing Binary (NEB) Analysis

The dmag–RMS diagnostic plot (Figure 3) was used to check for possible NEB contamination. In this case, T1 (the target star) was not cleared, lying above the green and pink thresholds. This suggests the candidate could be associated with an eclipsing binary; however, exposure time and observing conditions likely contributed to the elevated RMS values. Improved observing conditions and longer exposures would be expected to reduce the RMS and potentially clear the target.

3.6 Final Light Curve

In AstroImageJ, the flux of the target star was compared against the combined flux of selected reference stars to generate differential photometry. This process corrects for variations introduced by atmospheric transparency, airmass, and instrumental noise, producing a table of relative flux as a function of time. Using these measurements, we constructed the final ground-based light curve of TOI-5516 [6,7]. The normalized flux was plotted alongside the predicted transit model (Figure 4), This allowed for a direct comparison between the expected TESS signal and our ground-based observations. The resulting curve provides critical insight into whether a transit event is consistent with the predicted ephemeris. Even though the overall scatter of our dataset was comparable to the expected transit depth, the light curve enabled us to test the alignment of dips in brightness with the predicted ingress and egress times. This step is essential for distinguishing genuine transit signals from random noise or systematics.

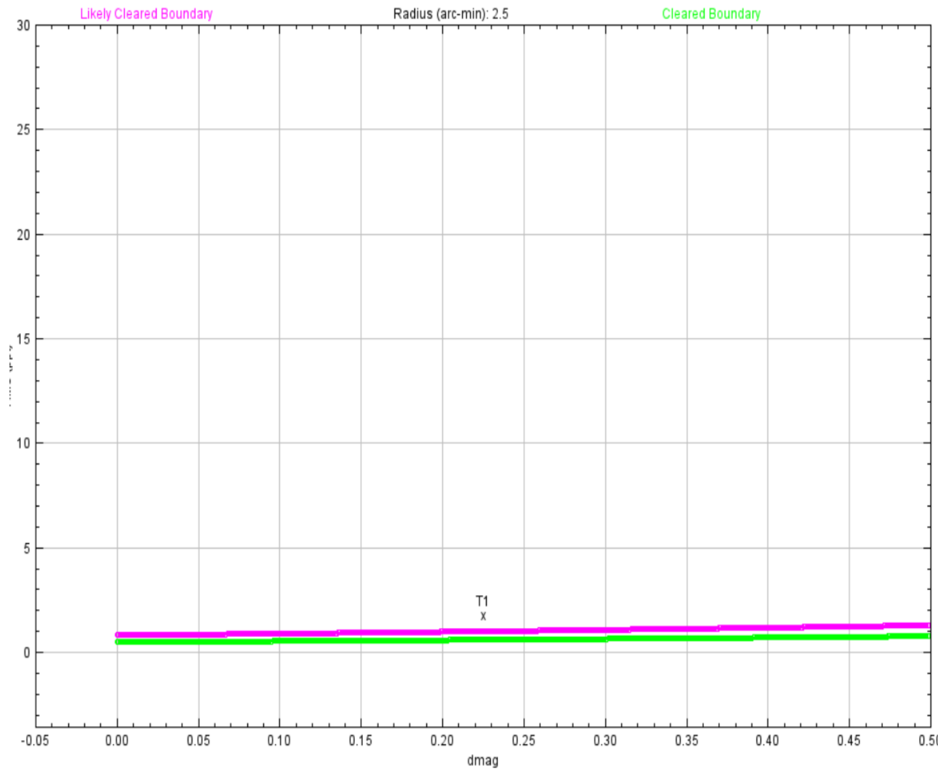


Figure 3. Dmag versus RMS diagnostic plot for TOI-5516.01 generated in *AstroImageJ*. The green and magenta curves represent the "cleared" and "likely cleared" boundaries, respectively. The target star (T1) falls above these thresholds, indicating that the photometric scatter is too high for a conclusive transit detection and that additional, higher-precision observations are required.

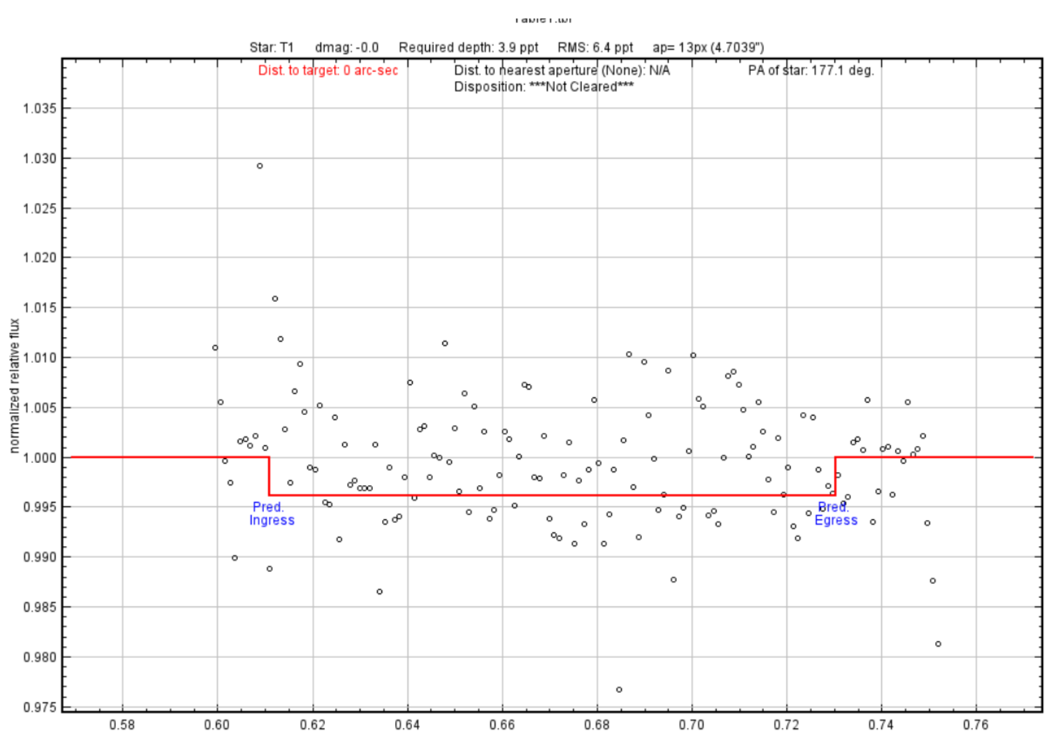


Figure 4. Ground-based light curve of TOI-5516.01 generated with AstroImageJ..

4. Results

The final ground-based light curve was constructed by plotting the relative flux of TOI-5516 alongside three reference stars, residuals with error bars, and additional parameters including Sky/Pixel_T1, Width_T1, AIRMASS, total counts, and X(FITS)/Y(FITS) (Figure 5).

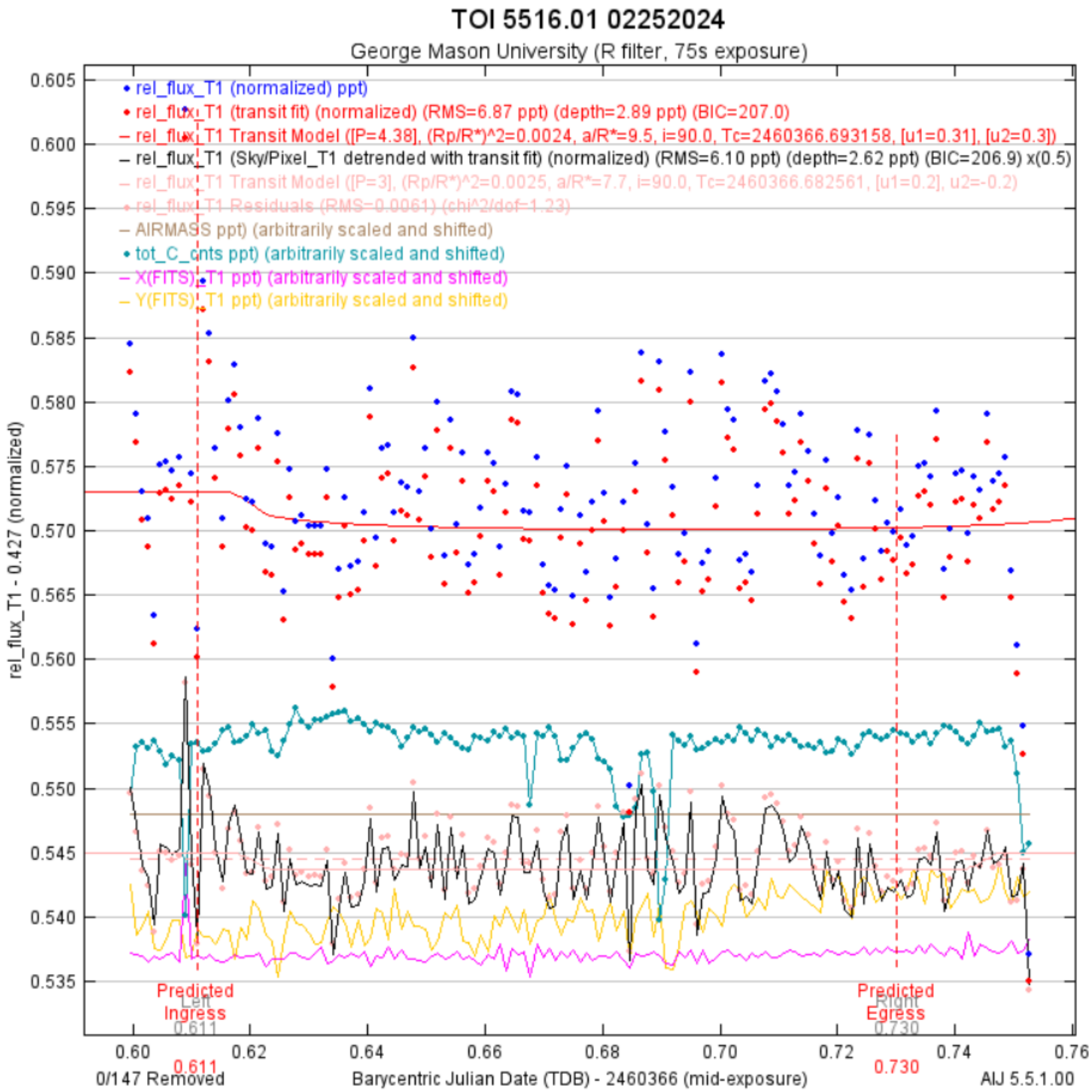


Figure 5. Ground-based light curve of TOI-5516.01 observed with the GMU 0.8 m telescope (R filter, 75 s exposures). Blue points show the normalized relative flux of the target star, and red points represent the transit fit. The solid red line indicates the predicted TESS transit model (period = 4.38 d), with vertical dashed lines marking the expected ingress and egress times. The fitted transit depth (~2.9 ppt) is smaller than the photometric scatter (RMS ~6.9 ppt), making the detection inconclusive. Additional trends from airmass, total counts, and x/y position (colored curves) are shown to illustrate systematics.

The differential light curve exhibited a transit event at the predicted time. Data scaling and shifting followed the TFOP SG1 guidelines (p. 8). The resulting curve enabled direct comparison between the TESS-predicted transit and our ground-based observations. AIJ overlays the predicted model curve (red line in the figure).It marks the Predicted Ingress and Egress times (blue labels). The fit estimates the transit depth and compares it with the measured scatter (RMS). By visually and quantitatively comparing the observed data to the TESS model, we can evaluate the plausibility of detecting the candidate transit within the limitations of our ground-based measurements. However, noise in the comparison star data and atmospheric variability limited the depth precision. Despite scatter, the transit depth was consistent with expectations for TOI-5516.01.

5. Discussion

Our analysis demonstrates the challenges of extracting high-precision light curves from modest ground-based telescopes. While we successfully detected a transit consistent with TOI-5516.01, significant scatter remained, preventing robust parameter refinement. These results illustrate both the promise and limitations of AIJ multi-aperture methods under less-than-ideal observing conditions.

The observational results for TOI-5516.01 remain inconclusive, as the diagnostic plots and light curves do not provide sufficient evidence to confirm a planetary transit. The Dmag versus RMS analysis shows that the target star lies above the cleared boundaries, meaning the scatter in our data is too large to rule out contamination from nearby eclipsing binaries or variability. This high noise level is also reflected in the light curve, where the fitted transit depth of ~6 ppt is nearly indistinguishable from the RMS scatter of ~5.4–6.4 ppt. Because the measured depth is not significantly larger than the noise, any transit signal is hidden within the uncertainties of the dataset. While the predicted ingress and egress times align with slight dips in the light curve, the signal-to-noise ratio is too low to consider the detection statistically meaningful.

This inconclusive outcome also complicates classification. Hot Jupiters are typically defined by both short orbital periods and high equilibrium temperatures. TOI-5516.01 meets the first criterion, with a period of ~1.389 days well within the hot Jupiter regime, but its equilibrium temperature of ~1281 K is significantly cooler than the ~2000–3000 K range usually associated with this class. As a result, even if a planet is present, it does not fit neatly into the hot Jupiter category [8].

Overall, the figures presented illustrate the limitations of the current dataset: the NEB check does not clear the target, the measured transit depth is not distinguishable from noise, and the stellar and orbital properties leave the candidate's classification uncertain. TOI-5516.01 therefore remains an intriguing but unconfirmed object. Further work, such as higher-precision photometry, repeated monitoring of multiple transit events, radial velocity measurements, and false positive probability analyses—will be necessary to determine whether this candidate truly hosts a planet.

6. Conclusion and Future Work

Our ground-based observations of TOI-5516.01 were not precise enough to clearly detect a transit. Although the timing of our data lined up with when a transit was predicted, the expected signal was very small and difficult to separate from the natural scatter in the measurements. Because of this, we cannot confirm or rule out the possibility that a transit occurred.

The possible classification of TOI-5516.01 as a Hot Jupiter is also unclear. Its orbital period of about 4.38 days matches what is usually seen in hot Jupiters, but its estimated temperature (~1281 K) is cooler than most planets in this category [13]. This means the planet, if it exists, does not fit neatly into the typical definition of a hot Jupiter.

To better understand TOI-5516.01, several directions for future work are recommended:

- Higher-precision photometry with larger ground-based telescopes or space telescopes to reduce scatter and confirm the shallow signal.
- Radial velocity (RV) measurements to test whether the star's motion reveals the presence of a massive companion.
- False positive probability (FPP) analyses to rule out scenarios like eclipsing binaries or background stars.
- Longer monitoring campaigns to capture multiple predicted transit events, improving confidence in the signal.
- Spectroscopic follow-up to study the host star more carefully, refine its parameters, and check for stellar activity that could mimic transits.
- Global modeling with tools such as ExoFASTv2, which allow simultaneous fitting of photometric and spectroscopic data for more robust results.

In summary, while our results are inconclusive, TOI-5516.01 remains an interesting target for further study. Confirming or disproving its planetary nature would improve our understanding of short-period gas giants and how they form.

Acknowledgements

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References

1. Astrometry.net. (2024). Astrometry.net. https://nova.astrometry.net/users/51752
2. Collins, K. A., Kielkopf, J. F., Stassun, K. G., & Hessman, F. V. (2017). AstroImageJ: Image processing and photometric extraction for ultra-precise astronomical light curves. The Astronomical Journal, 153(2), 77. https://doi.org/10.3847/1538-3881/153/2/77
3. Davoudi, F., Jafarzadeh, S., Poro, A., Basturk, O., Mesforoush, S., Fasihi Harandi, A., Gozarandi, M., Zare Mehrjardi, Z., Maley, P., Khakpash, S., Rokni, K., & Sarostad, A. (2020). Light curve analysis of ground-based data from exoplanets transit database. New Astronomy, 76, 101305. https://doi.org/10.1016/j.newast.2019.101305
4. Exoplanets | Center for Astrophysics. (n.d.). CfA Harvard. https://www.cfa.harvard.edu/research/topic/exoplanets

5. Exoplanets | MIT Kavli Institute. (2018, August 13). MIT Kavli Institute for Astrophysics and Space Research. https://space.mit.edu/research/exoplanets/
6. Freudling, W., Romaniello, M., Bramich, D. M., Ballester, P., Forchi, V., García-Dabló, C. E., Moehler, S., & Neeser, M. J. (2013). Automated data reduction workflows for astronomy. Astronomy & Astrophysics, 559, A96. https://doi.org/10.1051/0004-6361/201322494
7. Guerrero, N. M., Seager, S., Huang, C. X., Vanderburg, A., Garcia Soto, A., Mireles, I., Hesse, K., et al. (2021). The TESS objects of interest catalog from the TESS prime mission. arXiv preprint arXiv:2103.12538. https://arxiv.org/abs/2103.12538
8. Hot Jupiter - NASA Science. (2008, February 29). NASA Science. https://science.nasa.gov/resource/hot-jupiter/
9. Mantovan, G., Montalto, M., Piotto, G., Wilson, T. G., Collier Cameron, A., Majidi, F. Z., Borsato, L., Granata, V., & Nascimbeni, V. (2022). Validation of TESS exoplanet candidates orbiting solar analogues in the all-sky PLATO input catalogue. Monthly Notices of the Royal Astronomical Society, 516(3), 4432–4447. https://doi.org/10.1093/mnras/stac2451
10. NASA. (2024). TESS ground-based follow-up program. https://asd.gsfc.nasa.gov/archive/tess/ground_based_followup.html
11. Reefe, M., et al. (2022). Asynchronous object-oriented approach to the automation of the 0.8-meter George Mason University campus telescope in Python. Journal of Astronomical Telescopes, Instruments, and Systems, 8(2), 027002. https://doi.org/10.1117/1.JATIS.8.2.027002
12. TOI-5516 | NASA Exoplanet Archive. (2022). Caltech/NASA Exoplanet Archive. https://exoplanetarchive.ipac.caltech.edu/overview/TOI-5516.01#planet_TOI-5516-01_collapsible
13. Wang, J., Fischer, D. A., Horch, E. P., & Huang, X. (2015). On the occurrence rate of hot Jupiters in different stellar environments. The Astrophysical Journal, 799(2), 229. https://doi.org/10.1088/0004-637X/799/2/229
14. What's a transit? - NASA Science. (n.d.). NASA Science. https://science.nasa.gov/exoplanets/whats-a-transit/
15. YouTube. (2017, January 30). AstroImageJ (AIJ): Processing exoplanet transits (start to finish) [Video]. YouTube. https://www.youtube.com/watch?v=GW--rE5O-c8